Fiscal Year:	FY 2010	**Task Last Updated:**	FY 01/14/2011
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Project Title:	Effects of Space Radiation on Degenerative Tissue Disease, Genetic Instability and Oxidative DNA Damage in Ataxia Telangiectasis Deficient Mice		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	90095-8347	**Congressional District:**	33
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 Radiation Biology NNH04ZUU005N
Start Date:	08/15/2005	**End Date:**	08/14/2010
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 8/14/2010 per A. Chu/ARC (8/09) NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes then, per JSC/A. Chu-ARC (jvp 4/2009)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Fleming, Sheila (University of California, Los Angeles)		
Grant/Contract No.:	NNA05CV34G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Atm (ataxia telangiectasia mutated) heterozygosity has a very high prevalence in the human population and seems to be a risk factor for cardiac damage and cancer. Radiation also causes reactive oxygen species, which also seem to be involved in the etiology of Ataxia Telangiectasia, cardiac damage, movement disorder and cancer. Oxidative DNA damage, DNA strand breaks and genetic instability may be involved in the etiology of these diseases. Atm deficient homozygous patients and mice show an elevated frequency of DNA deletions and signs of oxidative stress in their cells. Preliminary results show that DNA deletions at the pun locus are induced by ionizing radiation. The DNA deletion assay is based on the quantification of black spots on fur and eyes resulting from reversion of the pun mutation. This reversion occurs by deletion of 70 kb of an internal duplication within the p gene. Aim 1 is to determine whether AtmDSRI homozygous or heterozygous mice may show an increased frequency of space radiation induced DNA deletions. We also found that the elevated frequency of DNA deletions in Atm truncation deficient mice is completely reverted to wildtype levels by nutritional supplementation with the radical scavenger and glutathione precursor N-acetyl cysteine (NAC). Thus, we will test in aim 2 whether AtmDSRI homozygous or heterozygous mice nutritionally supplemented with NAC may show reduced frequencies of space radiation induced DNA deletions whereas mice supplemented with the prooxidant and inhibitor of glutathione metabolism, BSO may show an increased frequency of radiation induced deletions, which will be tested in aim 2. In aim 3 we will test whether AtmDSRI homozygous and heterozygous mice may show an increased frequency of space radiation induced cardiac damage or cancer. Aim 4 is to test whether AtmDSRI homozygous and heterozygous mice may show an increased frequency of space radiation induced ataxia. Implications for human health may include that Atm missense mutation heterozygosity (a significant proportion of the human population) may cause an increased risk for space radiation induced cardiac damage ataxia or cancer and that potential astronauts may benefit from being tested for Atm heterozygosity and that chemoprevention with antioxidants may reduce the space radiation induced genomic instability and possibly cancer in Atm heterozygous people.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Implications for human health may include that Atm missense mutation heterozygosity (a significant proportion of the human population) may cause an increased risk for space radiation induced cardiac damage, ataxia or cancer and that potential astronauts may benefit from being tested for Atm heterozygosity and that chemoprevention with antioxidants may reduce the space radiation induced genomic instability and possibly cancer in Atm heterozygous people.
Task Progress:	During our extension year, we followed-up on interesting and unexpected results from our proposed experiments. We found that a one-time exposure to 1 Gy, 1 GeV/n 56Fe ions decreased lifespan in wildtype mice but not Atm heterozygous or Atm-deficient mice. ATM is important in DNA repair, apoptosis, and cell cycle regulation. Atm-deficiency is almost synonymous with radiosensitivity. Therefore, it was interesting that 1 Gy, 1 GeV/n 56Fe ions did not significantly affect lifespan in Atm-deficient mice. Moreover, we found that while 2 Gy low-LET gamma rays caused an increase in DNA damage in Atm-deficient mice compared to wildtype mice after 6 hours, 1 Gy, 1 GeV/n 56Fe ions caused about the same amount of damage in both Atm-deficient and wildtype mice. Finally, we found that while cancer incidence was not significantly affected after exposure to 1 Gy accelerated iron ions, cancer latency was differentially affected depending on genotype. Cancer latency was shortened in Atm heterozygous mice, prolonged in Atm homozygous mice, and was unchanged in wildtype mice. The extended cancer latency in Atm homozygous mice after exposure to high-LET radiation was an unexpected result. Together, these results indicate that while Atm-deficient mice are radiosensitive, there may be some benefit in using high-LET radiation cancer therapy, which can theoretically deposit the majority of energy at the target site rather than normal, healthy tissue. In our extension year, we used beam time during one run to assess apoptosis after exposure to high- and low-LET radiation and at different doses, if there is a threshold in Atm-deficient mice at which cancer is no longer delayed, but accelerated by high-LET radiation, and if spontaneous cancer in Atm-deficient mice can be alleviated with high-LET radiation. We found that while2 and 6 Gy gamma rays did not affect apoptosis as measured by annexinV-positive cells after 6 hours in mice regardless of genotype, annexinV-positive cells were increased in Atm heterozygous and Atm-deficient mice after exposure to 2 Gy, 1 GeV/n 56Fe ions, and in wildtype mice 6 hours after exposure to 6 Gy, 1 GeV/n 56Fe ions. Although this is not the optimal time to look at apoptosis, we wanted to compare levels of apoptosis in the peripheral blood at the same time that we looked at DNA damage. These results indicate that while DNA damage is occurring in Atm-deficient mice after exposure to gamma rays, the cells are trying to repair the damage rather than undergo apoptosis whereas after exposure to high-LET radiation, cell death is more prevalent. These results may help to explain the extended cancer latency after exposure to high-LET radiation as precancerous cells may be undergoing apoptosis rather than DNA repair and/or misrepair and survival. This pattern, however, does not translate in Atm heterozygous mice where apoptosis is increased after 2 Gy, 1 GeV/n 56Fe ions but cancer latency is shortened. In Atm heterozygous mice, however, there may be a dose-dependent difference between 2 Gy and 1 Gy, at which the cancer latency studies were done. We also wanted to determine if there was a dose at which cancer latency would be decreased in Atm-deficient mice. We irradiated 2 mice with 4 Gy gamma rays and have not an effect on lifespan or carcinogenesis. Therefore, we elected to irradiate mice with 6 Gy gamma rays or 1 GeV/n 56Fe ions. At this dose, however, mice did not survive over 1 month post-radiation while wildtype mice are all currently alive about 4.5 months post-exposure. Therefore, Atm-deficient mice may be more sensitive to radiation-induced carcinogenesis at a dose between 4 and 6 Gy gamma rays and 2-6 Gy, 1 GeV/n 56Fe ions. Finally, to determine if spontaneous tumors could be improved after exposure to high-LET radiation, we scanned mice using PET/CT scans to locate tumors and then exposed 2 mice to 8 Gy, 600 MeV/n 56Fe ions using a 1 cm in diameter collimator. The collimator targeted exposure to the left shoulder of both mice. We found that after a 1-time exposure, the tumor shrunk substantially and the mouse seems to be doing well 4.5 months after exposure. In conclusion, although our results are preliminary, they provide evidence that radiotherapy using high-LET radiation may be a feasible option in radiosensitive patients such as ATM-deficient patients. Results from these studies have been incorporated into grant proposals and will be the basis for future studies.

Bibliography Type:	Description: (Last Updated: 08/25/2020)
Articles in Peer-reviewed Journals	Yamamoto ML, Hafer K, Reliene R, Fleming S, Kelly O, Hacke K, Schiestl RH. "Effects of 1 GeVnucleon (56)Fe particles on longevity, carcinogenesis, and neuromotor ability in Atm-deficient mice." Radiation Research, 2011 Feb;175(2):231-9. Epub 2010 Nov 23. PMID: 21268717 , Feb-2011
Dissertations and Theses	Yamamoto ML. "Gene-Environment Interactions: Roles of DNA repair proteins in preventing disease and injury." Dissertation, University of California, Los Angeles, November 2009. Published November 2010. , Nov-2010